

NURSING WITH LOLO

# Laboratory Values

Trend-based review of CBC, chemistry, electrolytes, renal clues, and critical communication.

Simple explanations - nursing priorities - memory cues - original practice - rationales

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Educational study aid - verify current instructor and facility requirements

# What you will be able to do

Use these objectives to guide active recall before and after the lesson.

## Learning objectives

- Compare findings of fluid-volume deficit and excess.
- Connect sodium, potassium, magnesium, and calcium changes with priority assessments.
- Recognize findings requiring cardiac, neurologic, or respiratory escalation.
- Use safe monitoring principles for ordered fluid and electrolyte therapy.
- Use a consistent sequence to interpret pH, carbon dioxide, and bicarbonate.
- Identify primary respiratory and metabolic patterns.

## How to use this guide

- Read the simple overview once.
- Close the guide and explain the priority in your own words.
- Answer the practice items before opening the rationale.
- Return to the lowest-confidence section tomorrow.

# The concept, made simple

Plain language first; precise nursing language follows.

## Core explanations

- Water moves between body spaces, while electrolytes help control nerves, muscles, rhythm, and fluid distribution.
- Sodium problems often show in the brain because sodium strongly affects water balance.
- Potassium problems often show in muscles and the heart because potassium affects electrical activity.
- A laboratory value never stands alone: symptoms, speed of change, kidney function, medicines, and the collection method all matter.
- pH describes acidity. PaCO<sub>2</sub> reflects the respiratory acid side; bicarbonate reflects the metabolic base side.
- In a primary respiratory problem, pH and PaCO<sub>2</sub> usually move in opposite directions. In a primary metabolic problem, pH and bicarbonate usually move together.

## Vocabulary move

- Name the body system or safety process.
- Connect the cue to the risk.
- State what the nurse should notice and do first.

# Assess before you act

A focused assessment organizes the data that make the next action safe.

## What to notice

- Trend weight, intake/output, vital signs, orthostatic symptoms, perfusion, edema, and lung sounds.
- Assess alertness, headache, seizure risk, strength, reflexes, cramps, and sensation.
- Review the electrolyte trend, kidney function, glucose, acid-base data, and collection quality.
- Review medicines, IV fluids, losses, nutrition, and recent procedures.
- Use ordered rhythm monitoring when potassium, magnesium, calcium, or symptoms create risk.
- Inspect every infusion for correct solution, route, rate, access condition, and pump settings.
- Assess airway, respiratory rate/depth/effort, alertness, circulation, and oxygenation before focusing on calculations.
- Confirm specimen type, collection time, current oxygen or ventilation support, and relevant treatment timing.

## Trend, do not snapshot

- Compare with baseline.
- Cluster related cues.
- Validate unexpected values.
- Escalate time-sensitive changes.

# Priority actions and safety boundaries

The safest answer protects the client while staying inside role, order, and policy.

## Priority actions

- For seizure, severe confusion, dangerous rhythm, marked weakness, or respiratory compromise, activate urgent response.
- Stop and verify an infusion when the solution, route, rate, access, or patient response is concerning.
- Report critical results according to policy and read back required information.
- Question an unexpected result and assess for specimen problems without dismissing possible illness.
- Avoid rapid independent correction; the safe pace depends on cause, duration, symptoms, and orders.
- For apnea, severe fatigue, reduced consciousness, shock signs, seizure, or dangerous rhythm, activate urgent response.
- Treat the patient's ABC threat before completing a detailed acid-base label.
- Report critical blood-gas results with specimen type, support settings, time, symptoms, and trend.

## Safety warnings

- Never give potassium by IV push, bolus, or direct injection; IV potassium must be diluted and infused with prescribed pump and monitoring safeguards.
- Rapid correction of some sodium disorders can cause severe neurologic harm.
- Fluid boluses or restriction can be harmful when applied without individualized assessment and orders.
- Do not dismiss a critical result solely because the patient has few symptoms.
- A near-normal pH can hide a mixed disorder or compensated serious problem.
- Do not interpret an ABG without knowing oxygen or ventilation support and specimen timing.
- Do not delay airway or perfusion support to calculate compensation.

# Memory that transfers

A memory aid is useful only when it leads back to clinical reasoning.

## Memory hooks

- Sodium: think water and brain.
- Potassium: think power for muscle and cardiac electricity.
- Volume trend: weight, output, pressure, pulse, lungs, and person.
- Respiratory opposite: pH and CO<sub>2</sub> move opposite in the primary disorder.
- Metabolic equal: pH and bicarbonate move the same direction.
- Patient first, pH second, cause third, trend always.

## Exam reset

- What can harm the client first?
- What data are missing?
- Is the client stable or unstable?
- Which action is least restrictive and within scope?

# Clinical judgment in motion

A patient receiving IV fluid develops new crackles and breathlessness. Weight has increased, urine output is falling, and the infusion is still running.

## Notice

- New respiratory symptoms
- Crackles
- Weight gain
- Falling output
- Active infusion

## Interpret

- The patient may no longer tolerate the current fluid plan.
- Continuing without reassessment could worsen fluid excess.

## Respond

- Assess ABCs, vital signs, saturation, lung sounds, edema, infusion details, and fluid trend.
- Pause or adjust only as authorized by policy/order and promptly notify the appropriate clinician.
- Carry out revised orders and reassess breathing, output, weight, and labs.

## Reflect

- Which trend shows why an originally appropriate fluid order may later require review?

# Practice before the rationale

Choose the best answer using the cue-risk-action sequence.

## Question 1

Which finding in a patient with an electrolyte disorder requires the fastest response?

- A. Request for a diet handout
- B. New seizure activity
- C. Question about tomorrow's lab time
- D. Mild chronic thirst at baseline

## Question 2

Which findings support fluid-volume excess? Select all that apply.

- A. Rapid weight gain
- B. New crackles
- C. Increasing edema
- D. Worsening orthopnea
- E. Improved breathing with stable weight



# Answers, rationales, and printable review

## Question 1: B

A seizure threatens airway and neurologic safety and requires emergency response before routine teaching or scheduling.

**Exam clue: Acute ABC and neurologic threats come first.**

## Question 2: A, B, C, D

Weight gain, crackles, edema, and orthopnea form a volume-excess pattern. Improved breathing and stable weight do not.

**Exam clue: Choose findings that all show accumulation or congestion.**

## Five-point review

- Fluid status is judged from trends, not one sign.
- Sodium disturbances commonly affect neurologic status; potassium disturbances can threaten rhythm.
- Kidney function and medicines shape electrolyte risk.
- Replacement and correction require the correct route, rate, monitoring, and follow-up.
- Assess the patient, verify unexpected data, and escalate critical change.

## References

MedlinePlus, U.S. National Library of Medicine: Vital signs - <https://medlineplus.gov/ency/article/002341.htm>

MedlinePlus, U.S. National Library of Medicine: Electrolyte Panel - <https://medlineplus.gov/lab-tests/electrolyte-panel/>

U.S. Food and Drug Administration: Medication Errors Related to CDER-Regulated Drug Products -

<https://www.fda.gov/drugs/drug-safety-and-availability/medication-errors-related-cder-regulated-drug-products>

National Council of State Boards of Nursing: 2026 NCLEX-PN Test Plan - <https://www.ncsbn.org/publications/2026-nclex-pn-test-plan>

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